

SOC BLIND SPOT

Automated Detection Coverage & Telemetry Gap Assessment Platform

Complete Project Report

Wazuh + Windows 11 Laboratory

Cybersecurity Portfolio Project

1. Executive Summary

This project evaluates security monitoring coverage in a controlled Windows 11 laboratory using Wazuh. The goal is to identify telemetry and detection blind spots through controlled event generation, endpoint verification, Wazuh Threat Hunting, and Security Configuration Assessment (SCA).

The assessment successfully validated authentication, PowerShell, application-error, Windows Defender, process-creation, account-creation, and scheduled-task visibility. Windows Firewall events were confirmed locally, and the Wazuh agent was configured to analyze the Firewall event channel, but the controlled Firewall test did not appear in Threat Hunting. This is the principal confirmed blind spot.

2. Objectives

- Deploy and validate Wazuh as the SOC monitoring platform.
- Connect and validate a Windows 11 endpoint.
- Assess and improve Windows security configuration using Wazuh SCA.
- Validate detection coverage using controlled Windows events.
- Identify telemetry gaps and document them as SOC blind spots.
- Produce an evidence-based detection coverage assessment.

3. Laboratory Environment

4. Deployment and Validation

The Wazuh single-node stack was deployed successfully. The Windows endpoint was enrolled as agent 001 and reported Active. Network connectivity between the Windows endpoint and Wazuh manager was validated, and Windows events appeared in Wazuh Threat Hunting.

5. Security Configuration Assessment

The CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0 was evaluated through Wazuh SCA. The latest recorded result was 145 passed, 328 failed, and 9 not applicable, for approximately 30% compliance.

Because the endpoint is Windows 11 Home while the benchmark is for Enterprise, the SCA score must be interpreted with edition limitations in mind. SCA compliance is a configuration metric and is not the same as detection coverage or overall project completion.

6. Hardening Performed

7. Detection Coverage Method

Each use case was tested by generating controlled activity, checking Windows locally when necessary, and searching Wazuh Threat Hunting for the corresponding event/rule. This method distinguishes endpoint event generation from actual SOC visibility.

8. Detection Coverage Matrix

9. Key Findings

- Successful and failed Windows authentication activity was visible in Wazuh.

- PowerShell Operational telemetry was initially absent, then became visible after the event channel was explicitly configured.
- Windows Application Error telemetry was successfully detected by Wazuh (Rule 66002).
- Windows Defender telemetry was successfully detected (including scan activity, Rule 62108).
- Process Creation Event ID 4688 was successfully detected (Rule 67027).
- Account creation and scheduled-task detection tests were completed and recorded as detected.
- Windows Firewall events were verified on the endpoint and the Wazuh agent log confirmed that the Firewall event channel was analyzed, but the controlled firewall test did not appear in Threat Hunting. This is the primary confirmed blind spot.

10. Blind-Spot Analysis

The Firewall finding demonstrates the central principle of this project: an event can exist on an endpoint without being visible to the SOC. The Windows Firewall log generated events and the Wazuh agent was configured to analyze the event channel, yet the controlled test did not produce a corresponding Threat Hunting result. Further investigation should focus on event-channel ingestion, filtering, manager processing, and rule mapping.

11. Coverage Metrics

The final test matrix contains 9 documented use cases. Eight were recorded as detected and one as a blind spot, giving an observed test-case detection coverage of $8/9 = 88.9\%$. This is a project test metric only; it is not a claim that Wazuh detects 88.9% of all possible Windows threats.

Current overall project completion is approximately 80%. The remaining work is primarily evidence organization, formal findings, recommendations, and final presentation/report packaging.

12. Recommendations

- Investigate the Windows Firewall ingestion/detection gap using Wazuh agent debug logging and event-channel validation.
- Maintain a telemetry-to-use-case matrix for every important Windows security log.
- Retest detection coverage after configuration changes or Wazuh upgrades.
- Keep SCA compliance and detection coverage as separate project metrics.
- Document Windows Home-versus-Enterprise benchmark limitations.
- Add SOC triage and response playbooks for production deployment.

13. Evidence Register

Evidence captured during the implementation includes Wazuh agent Active status, network connectivity, SCA results, successful/failed logon detections, PowerShell telemetry, Windows Application Error detection, Windows Defender detection, process creation detection, and local Firewall events plus the Wazuh Firewall-channel analysis log.

Recommended screenshot names:

- 01_Wazuh_Agent_Active.png
- 02_SCA_Latest_30_Percent.png
- 03_Successful_Logon_4624.png
- 04_Failed_Logon_4625.png

- 05_PowerShell_Telemetry.png
- 06_Application_Error_Rule_66002.png
- 07_Windows_Defender_Rule_62108.png
- 08_Process_Creation_4688_Rule_67027.png
- 09_Firewall_Local_Event.png
- 10_Firewall_Channel_Loaded.png
- 11_Firewall_Blind_Spot_No_Wazuh_Result.png

14. Conclusion

The SOC Blind Spot project successfully demonstrates a repeatable method for measuring whether Windows security telemetry actually reaches a SOC platform and produces detections. Wazuh was deployed and integrated with Windows 11, multiple important telemetry sources were validated, selected SCA controls were hardened, and a concrete Firewall telemetry blind spot was identified. The project therefore demonstrates detection engineering and coverage assessment rather than simply SIEM installation.